

First Stable Index

authored by Arpit Raj , LNMIIT Jaipur , September 4, 2026

Solution Code

```
class Solution {
public:
    int firstStableIndex(vector<int>& nums, int k) {

        int n = nums.size();
        for(int i = 0; i < n; i++) {

            int maxu = *max_element(nums.begin(), nums.begin() + i + 1);
            int minu = *min_element(nums.begin() + i, nums.end());

            if(maxu - minu <= k) return i;
        }

        return -1;
    }
};
```

Logic & Explanation

This approach systematically evaluates each possible array split to verify if it satisfies the stability condition. Here is a step-by-step breakdown of the logic:

- **Iteration:** The algorithm iterates through every possible index i from 0 to $n-1$.

- **Prefix Maximum Check:** For a given index i , it calculates the maximum element present in the prefix array segment from index 0 up to i using the `max_element` function.
- **Suffix Minimum Check:** Similarly, it computes the minimum element present in the suffix array segment starting from index i up to the end of the array using `min_element`.
- **Stability Condition:** An index qualifies as a stable point if the computed difference between the maximum of the prefix (`maxu`) and the minimum of the suffix (`minu`) is less than or equal to our threshold k .
- **Result Delivery:** The loop instantly returns the very first index i that fulfills this stability criterion. If the loop completes without finding any such index, it defaults to returning -1 .

Implementation Note: The current code is written clearly to reflect the problem's mathematical definition. Calculating the maximum and minimum sub-arrays iteratively inside the loop gives a direct simulation of the problem statement.